



Algorithmic indirect discrimination, fairness and harm

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Received: 9 December 2022 / Accepted: 18 July 2023

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Abstract

Over the past decade, scholars, institutions, and activists have voiced strong concerns about the potential of automated decision systems to indirectly discriminate against vulnerable groups. This article analyzes the ethics of algorithmic indirect discrimination, and argues that we can explain what is morally bad about such discrimination by reference to the fact that it causes harm. The article first sketches certain elements of the technical and conceptual background, including definitions of direct and indirect algorithmic discrimination. It next introduces three prominent accounts of fairness as potential explanations of the badness of algorithmic indirect discrimination, but argues that all three are vulnerable to powerful leveling-down-style objections. Instead, the article demonstrates how proper attention to the way differences in decision scenarios affect the distribution of harms can help us account for intuitions in prominent cases. Finally, the article considers a potential objection based on the fact that certain forms of algorithmic indirect discrimination appear to distribute rather than cause harm, and notes that we can explain how such distributions cause harm by attending to differences in individual and group vulnerability.

Keywords Algorithmic bias · Fairness · Discrimination · Automated decision-making · Harm · Indirect discrimination

1 Introduction

Over the past decade, it has become increasingly more common that important decisions about issues, such as banking, health care, social services, and criminal justice, are made, not by a human decision-maker, but by an automated decision system (ADS) employing algorithmic profiling. At a very general level, such systems work roughly as follows: a software algorithm is fed data about you, performs a statistical analysis of these data to calculate the probability that you do or do not possess a target property, and renders or recommends a decision based on whether that probability is above or below a specific threshold.

ADS potentially offers more efficient and precise evaluation of your case than a human assessment because it can rely on vast amounts of data, apply mathematically sophisticated analyses precisely and consistently, and avoid human cognitive biases [4, 60, 110]. The advent of ADS has not, however, been greeted with unanimous enthusiasm. Scholars, institutions, and activists have voiced strong concerns about the use of such systems (e.g., [1, 36, 37, 39, 53, 56,

82, 82–85, 86, 90, 92, 111]). One of the most prominent such concerns is that ADS can discriminate against vulnerable groups even when discrimination is neither intended nor coded into the ADS. I shall refer to this as indirect algorithmic discrimination (and clarify shortly).

Despite the critical attention it has drawn in public debate, it is only recently that moral philosophers have begun to analyze the ethics of indirect algorithmic discrimination in light of the methods and theories of applied ethics and established results within the ethics of discrimination.¹ The central ambition of this paper is to contribute to these efforts by exploring algorithmic indirect discrimination from a philosophical perspective, and by advancing a plausible account of what makes algorithmic indirect discrimination morally bad.

¹ Some recent contributions include [15, 19, 38, 40, 47, 49, 74, 77, 103]. Legal scholars have engaged more extensively with the issue. (See e.g., [2, 9, 20, 31, 51, 55, 61, 93, 109].) By far the most concerted focus has come from data scientists and computer scientists, particularly within the so-called “fair machine learning” community. Some central contributions include [10, 22, 26, 33, 42, 44, 46, 59, 65, 24.] For a good, recent overview, see: [17].

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The account I will defend holds that indirect algorithmic discrimination is morally bad because it causes harm. Perhaps there are further reasons why indirect discrimination is morally bad, but the harm-based account is both very plausible and capable of explaining our intuitions about important cases. Crucially, however, I will argue that algorithmic discrimination is never morally bad because it fails to meet certain parity conditions for the distribution of outcomes that have been the center of much attention in the fair machine learning literature.

The argument proceeds as follows: Sect. 2 presents a conception of discrimination, particularly as regards the distinction between direct and indirect discrimination. Section 3 introduces classification, accuracy and error type as possible measures of unequal effects, and three corresponding parity conditions that have been widely discussed in the literature on fairness in machine learning. Section 4 presents a series of leveling down-type objections to argue that none of the parity conditions are plausible accounts of what is morally bad about indirect algorithmic discrimination. Section 5 introduces the harm-based account and notes how it can explain our intuitions once we take proper account of important differences between decision scenarios. Section 6 presents a challenge to the harm-based account, that it might be unable to explain the moral badness of scenarios where harmful classifications are unavoidably distributed. Section 7 answers the challenge and further details the harm-based account by pointing out how differences in individual and group vulnerabilities affect the harms produced by classification. Jointly, sections five to seven demonstrate that we can explain the moral badness of indirect algorithmic discrimination in relevant cases by reference to the fact that the ADS causes harm. Section 8 concludes by briefly considering some implications and limitations of the argument.

Let me make two notes of clarification before we begin. I shall speak throughout of discrimination being “morally bad”. By an act, practice or ADS being morally bad, I mean that it is a wrong-making feature of that thing, which is to say that it grounds a *pro tanto* reason against performing the act, sustaining the practice or using the ADS. I thus leave open the possibility that there could be situations where morally bad discrimination is all-things-considered permissible.

Furthermore, I shall speak throughout of “majority groups” and “majority persons” versus “minority groups” and “minority persons”. I use these terms as placeholders for the groups we might be concerned with in the context of algorithmic indirect discrimination. In practice it will often be important which specific groups are at stake—I discuss two reasons why in Sect. 7—but for many general points it will be helpful to consider discrimination of persons and groups in the abstract.

2 What is algorithmic indirect discrimination?

The focus of this article is the moral badness of indirect algorithmic discrimination. One difficulty for any such analysis is that discrimination is a complex phenomenon.² In this section, I define and distinguish between direct and indirect discrimination, and briefly sketch how algorithmic indirect discrimination can occur.

What are we to understand by “discrimination”, then? For the purposes of this article, I shall say that:

Direct discrimination. An agent *directly* discriminates against minority persons *iff* (i) she treats minority persons differently than she treats or would treat majority persons, (ii) her treatment of minority persons is disadvantageous, (iii) the difference in treatment is caused by minority persons being minority persons, and (iv) the differential treatment is in some respect morally bad.

There are different ways of defining discrimination. This definition reflects one general approach in the literature, but not the only possible approach (cf. [67, 70, 103, 104]). A few points may be worth clarifying.

First, the definition I adopt here is moralized, such that an act is discrimination only if it is in a relevant respect morally bad. I am not persuaded that this is generally speaking the most useful or accurate way to define discrimination, but for present purposes it will make the discussion easier (cf. [34], [70], [105]).

The definition is also generic, such that it does not restrict discrimination to a particular set of groups. On this point, I do think that a generic definition, as opposed to a definition that restricts discrimination to a particular set of groups, is generally more useful. In this context, however, the definition is generic for purely pragmatic reasons. I use “minority person” and “majority person” as placeholders, and the analysis is compatible with both a generic and a narrow conception of the groups that can be filled in. [99].

Finally, it is worth noting the way the definition distinguishes between equal treatment and equal effect. For treatment to be equal, it must be capable of being suitably described as performing the same act. For the effect to be equal, it must be the case that it makes neither group better or worse off. A chatbot offering assistance in English to

² Over the past decade-and-a-half (roughly) there has been increased philosophical interest in discrimination, which the current analysis draws on. Prominent contributions include [23, 34, 50, 52, 57, 70, 72, 73, 83].

both English-speakers and non-English-speakers meets the former but not the latter condition. A gym requiring men to change in locker room A and women to change in locker room B meets the latter but not the former condition (supposing that the locker rooms are in other respects equally good).³

On this account, an ADS differentially treats a group if it employs membership of that group as a variable in its function, such that prediction of the target property depends on the value of the variable (member/non-member). When this is the case, group membership causes differential treatment in the shape of differences in processing. For example, an ADS that increases the predicted market value of a house when the seller is male *because* a parameter in its function gives weight to gender treats men and women differently. If the differential treatment is also both disadvantageous to women and morally bad, then the ADS directly discriminates against women.

The focus of this article, however, is on algorithmic indirect discrimination. Building upon the above definition, how can we define it? I want to say that what makes the above a definition of specifically *direct* discrimination is the differential treatment at stake in conditions (i), and (iii). On this interpretation, we can modify these conditions to define indirect discrimination in the following way:

Indirect discrimination. An agent *indirectly* discriminates against minority persons *iff* (i) she treats (in the acts at stake) minority persons *equally* to how she treats or would treat majority persons, (ii) the equal treatment is disadvantageous to minority persons, (iii) the difference in *effect* is caused by minority persons being minority persons, and (iv) the treatment is in some respect morally bad.

As with the definition of direct discrimination, there are other possible ways of defining indirect discrimination. Indeed, how best to define it is a matter of some controversy in the literature (cf. [5, 27, 34, 58, 70, 100]). Regardless of what one thinks is the best general account of indirect discrimination, the definition I adopt here is will be useful as a way of drawing out a distinction between two intuitively different ways an ADS can discriminate.⁴

In line with the distinction between treatment and effect set out above, the notion of equal treatment at stake here is

purely formal. An ADS treats two groups equally in this sense so long as its function does not employ a variable that corresponds to group membership to predict the target property. Thus, an ADS equally treats but *indirectly differentially affects* a group if it employs at least one variable that correlates with membership of the group, such that it has a tendency to predict different values for members of the group. For example, an ADS that employs “height” as a variable will thereby indirectly affect men and women differently because height correlates with gender. If the difference in effect is disadvantageous to one of the two groups, and morally bad, then the ADS indirectly discriminates against that group.

It is important to bear in mind in this context, that an ADS can indirectly differentially affect groups both when the correlations at stake are and are not spurious. [25, 81]. In the former case, the ADS can be trained on biased data, in which case it will tend toward affecting groups differently because it assigns inordinate weight to the relevant variable(s), or it can be fed biased data when employed, in which case even an ADS that assigns the appropriate weights will differentially affect groups. A common example is the uneven registration of criminal activity caused by discriminatory police patrolling, which may inflate the perceived criminality of over-patrolled neighborhoods and their residents [35, 41, 89].

An ADS can also indirectly differently affect groups based on *non-spurious* correlations between variables, groups and targets. If, to continue our (hopefully) innocent example from above, height correlates with both prowess in the sport of basketball and gender, then an ADS that attempts to predict basketball prowess and employs height as a variable will indirectly differentially affect men and women, because it tends to (accurately) predict higher scores for men than for women. In many less innocent cases, such correlations may themselves be the result of both historical and ongoing discrimination of these groups.

Finally, it is worth noting that ADS often employ hundreds, thousands or even millions of variables. At least in part as a result of the enormous amounts of data used, correlations can exist without developers or users being aware of them, such that indirect differential effect can and likely often will occur inadvertently. In light of these facts, we urgently need a way to identify those cases where an ADS is indirectly discriminatory, which is to say we need an account of what makes algorithmic indirect, disadvantageous differential effect morally bad (when it is). It is to this issue, that we now turn.

³ For general discussion of the difference between equal and differential treatment, and disadvantageous and neutral treatment, see: Lipert-Rasmussen, 2014, p.40ff, [100, 106]).

⁴ I have previously defended an account of indirect discrimination along these lines on general grounds, and argued that algorithmic discrimination in particular illustrates it as a generally plausible and useful way of distinguishing direct and indirect discrimination. See [100].

3 What might be bad about algorithmic indirect discrimination?

I have suggested above that an ADS indirectly differentially affects groups when it employs variables that correlate both with the target property and group membership. The result will be systematic differences in outcomes across groups. When does this amount to indirect discrimination?

Discussion of this issue within the data science and machine learning literature has mostly been conducted under the heading of “fairness” [10, 14, 22, 33, 45, 62]. “Fairness”, however, appears to often be used as a placeholder for ethical concerns more broadly. Thus, I shall assume that we are interested in explaining the moral badness of algorithmic indirect discrimination, regardless of whether this explanation happens to turn on a philosophically precise conception of fairness.⁵

Scholars within the data science and machine learning community have developed a range of accounts of what it means for an ADS to be fair. (For overviews, see [10, 14, 17, 22, 42, 81, 80].) Here I want to focus on three measures of distributions across groups: classification, accuracy, and error type. We focus on these both because these are measures that matter *to the person* (we return to this point below), and because they are representative or even the paradigmatic exemplars of three general approaches to fairness measures. Let me define each of the three in turn.

Classification measures an ADS’ distribution of *positive and negative* classifications.

A classification is positive, if the ADS predicts that the target has the property at stake, and negative if not. ADS might for example attempt to predict whether a loan applicant will default.⁶ Differential effect in terms of the distribution of classifications means unequal rates of positive to negative classifications. To measure classification, we can ask: what is the probability that a randomly selected person from the relevant group will be positively classified? Subsequently, we can compare these scores for minority persons and majority persons. If the scores are significantly

different, then the ADS differentially affects the two groups with respect to classification.⁷

Accuracy measures an ADS’ distribution of *true and false* classifications.

Differential effect in terms of accuracy means unequal rates of true to false classifications. To measure accuracy, we can ask: what is the probability that a randomly selected person from the relevant group is correctly classified? Subsequently, we can compare these scores for minority persons and majority persons. If the scores are significantly different, then the ADS differentially affects the two groups with respect to accuracy.

There are other ways of measuring accuracy, e.g., by specifying it for various sub-groups. It is often important, for example, to distinguish between accuracy measures that condition on classification and accuracy measures that condition on ground truth. We can, for example, measure accuracy for those positively/negatively classified by the model [positive predictive value (PPV) and negative predictive value (NPV), respectively] or for those who do or do not in fact possess the target property (sensitivity/recall/true positive rate, and specificity/selectivity/true negative rate, respectively) [81]. These different types of accuracy measurement will often be important in practice because predictive value, which conditions on classification, is more useful when a decision-maker needs to determine how much evidentiary weight to attribute to the model’s prediction in a specific case, while true rates, which condition on ground truth, are more useful as an estimate of the odds a specific person has of receiving a correct prediction (cf. [51]). Much of the discussion around fairness has focused on these more specific accuracy measurements, including the celebrated proofs that in realistic scenarios models cannot simultaneously achieve parity in more than a subset of these specific accuracy measurements [21, 62], for discussion, see [14, 47, 48]. However, all of these measures ultimately concern the distribution of true/false predictions, and as such the leveling down objection I raise against parity of accuracy below applies to the variations. Here I focus on the stated measure of accuracy as arguably the simplest measure for illustrating this point.

Error type measures an ADS’ distribution of *false positive and false negative* classifications.

Differential effect in terms of error type means unequal rates of false positives to false negatives. To measure error type, we can ask: what is the probability that a randomly

⁵ There are theoretically precise conceptions of fairness in the philosophical literature, but they have not been widely employed in the debate on fair machine learning. Prominent examples include [16, 29, 84, 91]. See also [15].

⁶ For simplicity, I shall focus mostly on classification problems, where ADS attempts to predict the presence of a target property but the points transfer readily to regression problems, where ADS attempts to predict the value of the target property (“What is this person’s age?”).

⁷ “Significantly” because we must allow for minor differences attributable to randomness. Also, note that negative and positive classifications are symmetrical in the sense that increasing the probability of one equally reduces the probability of the other. Thus, we need only review one of the two to measure disparity.

selected person from the relevant group *who is misclassified* receives a false positive? Subsequently, we can compare these scores for minority persons and majority persons. If the scores are significantly different, then the ADS differentially affects the two groups with respect to error type.

Now, for each of the three measures, we might require that ADS affects minority persons and majority persons equally in the pertinent sense. Thus, we might require:

Parity of classifications (PC): Classification of the minority group is (roughly) equal to classification of the majority group.⁸

In the fair machine learning literature, what I here label “parity of classifications” is commonly referred to as “statistical parity”, “demographic parity”, or the requirement of “independence”.⁹ Parity of classifications (PC) ensures that classifications are distributed to minority persons in proportion to their fraction of the total population. Thus, if women are half the persons in a society, then PC requires that they will receive half of the positive and half of the negative classifications, while allowing, for example, that both minority and majority persons are more frequently classified negatively than positively. An often discussed application is representation in political and business leadership. Most persons will not obtain a position as a political or business leader, but we may consider it desirable that among the small group of persons who *do* become such leaders roughly half are men and roughly half women.

We might also require:

Parity of accuracy (PA): Accuracy for the minority group is (roughly) equal to accuracy for the majority group.

As noted above, there are other measures of accuracy, and a range of parity conditions associated with each of these. In the fair machine learning literature, the measure I employ here is also labeled “overall accuracy equality”.¹⁰

⁸ “Roughly” because, as before, we should presumably allow space for minor differences attributable to randomness.

⁹ The measure is conventionally formalized as $P[\hat{Y}=1 | S=0] = P[\hat{Y}=1 | S=1]$, where $\hat{Y}$ is the classification and S denotes group status, such that the condition requires that the probability of a positive classification conditional on majority group membership is equal to the probability of a positive classification conditional on minority group membership.

¹⁰ The measure is conventionally formalized as $P[\hat{Y}=y | Y=y \cap S=0] = P[\hat{Y}=y | Y=y \cap S=1]$, where $\hat{Y}$ is the classification, Y is the true status, y are the values that $\hat{Y}$ and Y can assume ($\{0,1\}$ in a binary classification problem) and S denotes group status, such that the condition requires that the probability of a classification being true conditional on majority group membership is equal to the probability of a classification being true conditional on minority group membership.

Parity of accuracy (PA) ensures that an ADS will be equally good at avoiding mistakes when classifying persons from the two groups. Thus, if Black, Asian, and Middle Eastern (BAME) persons make up one third of society, PA ensures that (roughly) a third of the errors the algorithm makes affect BAME persons. We might for example require that an ADS employed by a public child protection service to flag children in certain families as at risk, does not more or less frequently make mistakes when reviewing BAME families than when it reviews majority families.

Finally, we might require:

Parity of error type (PET): Error type for the minority group is (roughly) equal to error type for the majority group.

In the debate on fair machine learning, what I here label “parity of error type” is often also referred to as “treatment equality”, “equalized odds” or the requirement of “separation”.¹¹ Parity of error type (PET) ensures that the ADS will be equally likely to make the two kinds of mistake when classifying persons from the two groups.¹² Thus, if false positives make up half the errors when assessing theists, then PET requires that false positives make up (roughly) half the errors when assessing atheists. Notably, this is compatible with the ADS having different accuracy for the two groups—PET requires only that the ratios be equal for the errors actually made. In the most widely discussed example of applying this principle, if the criminal justice system relies on evaluations of offender risk of recidivism to determine bail or parole, then it makes a big difference to an offender whether a mistake is an over- or underestimation of her risk, and we might find it desirable that PET holds across relevant groups, such that e.g., misclassified BAME citizens are as likely to have received a false positive as misclassified White citizens (cf. [6, 30, 32, 66]).

¹¹ The measure is equivalent to the combination of the requirements of equal true- and false positive rates, which can be formalized as $P[\hat{Y}=1 | Y=1 \cap S=0] = P[\hat{Y}=1 | Y=1 \cap S=1] \cap P[\hat{Y}=1 | Y=0 \cap S=0] = P[\hat{Y}=1 | Y=0 \cap S=1]$, where $\hat{Y}$ is the classification, Y is the true status, such that the condition requires that the probability of a classification being positive conditional on true status being positive and majority group membership is equal to the probability of a classification being positive conditional on true status being positive and minority group membership and the probability of a classification being positive conditional on true status being negative and majority group membership is equal to the probability of a classification being positive conditional on true status being negative and minority group membership.

¹² As with accuracy above, we can specify more narrow conditions, such as parity of false positives (but not false negatives). However, as with the diverse accuracy measures noted above, the problems afflicting PET, discussed below, also apply to related parity conditions here.

In the remainder of this article, I will advance two central arguments. First, that failure to meet any or all of the parity conditions does not by itself make an ADS morally bad, and second, that the harm differential effects can cause *does* make an ADS morally bad, and is thus a plausible explanation of indirect algorithmic discrimination.

4 Against algorithmic parity conditions

Much of the discussion of indirect algorithmic discrimination has focused on the relative advantages and disadvantages of the parity conditions above and their various specifications. Famously, it has been proven that it is in realistic circumstances impossible to simultaneously satisfy certain sets of parity conditions, forcing developers to choose how to prioritize them [21, 62]. These impossibility results gives rise to a trilemma between the following three claims:

1. ADS that does not satisfy parity conditions $[P_1, P_2 \dots P_n]$ is morally bad.
2. ADS cannot (in real-life cases) simultaneously satisfy $[P_1, P_2 \dots P_n]$.
3. ADS is not (in real-life cases) unavoidably morally bad.

The three claims form the basis of a trilemma because each of the claims might seem attractive but they are jointly incompatible—at least one of them is false. As noted above, the second claim has been mathematically proven for certain sets of parity conditions, and the third is a very plausible ethical claim. Hence, the first claim must be false. One tempting response is to attempt to figure out which of the incompatible parity conditions to preserve and which to abandon. In this section, however, I will argue that PC, PA, and PET are each vulnerable to powerful leveling down-type objections.¹³ The answer to the question of which parity condition(s) we should preserve, because ADS must satisfy it in order to avoid indirect algorithmic discrimination, is “none”.

Let us consider first so-called parity of classification. PC, recall, obtains simply when the ratio of positive to negative classifications for minority persons is (roughly) equal to the ratio of positive to negative classifications for majority persons. PC has been widely criticized as a fairness measure because it is insensitive to the distribution of the target value (the objection I raise below illustrates this feature) [33, 46].

¹³ The original leveling down objection was famously raised by Derek Parfit against telic egalitarianism. [88] I say leveling down-type objections, because they are structurally similar but different in that they pertain to reasons, as opposed to values. (cf.[71], chapter 5).

It will nonetheless be helpful to consider the objection I raise against it below as an introduction to the objections raised against PD and PET. Consider therefore:

Medical. A hospital employs ADS to determine whether a particular form of cancer is malignant or benign. If untreated, malignant tumors are fatal, but treatment also has very bad side-effects.

In *Medical*, PC obtains between majority persons and minority persons if the proportion of majority patients who are diagnosed as having a malignant tumor is (roughly) equal to the proportion of minority patients who are diagnosed as having a malignant tumor. If, for example, one in five female patients is diagnosed as having a malignant tumor, then demographic parity between men and women obtains if one in five male patients are also diagnosed as having a malignant tumor. The question now is this: is there a moral reason to strive for demographic parity in ADS?

Although we often discuss PC as an important policy goal, e.g., in the context of representation of minority persons in political or business leadership positions, it seems clear that there is no moral reason to strive for PC in and of itself. Consider:

Medical 2. A hospital employs ADS to determine whether a particular form of cancer is malignant or benign. If untreated, malignant tumors are fatal, but treatment also has very bad side-effects. In majority persons, tumors are 50% likely to be benign and 50% likely to be malignant. In minority persons, tumors are 80% likely to be benign and 20% likely to be malignant.

In *Medical 2*, ADS with a 100% accuracy will assess far more majority persons than minority persons as having a malignant tumor (1:2 vs 1:5). This means that PC does not obtain (1:2 $\gg$ 1:5).¹⁴ It would be strange to say, however, that this disparity is in any way morally problematic. After all, it simply reflects the fact that tumors are more likely to be malignant in majority persons than in minority persons. In this case, we could achieve (or approach) PC only by misdiagnosing persons. This would be bad for these persons regardless of whether they were minority persons with malignant tumors misdiagnosed as healthy, or healthy majority persons misdiagnosed as in need of treatment. In either case, therefore, we would be “leveling down”, in the sense that we would be promoting a specific measure of equality (i.e., PC) by making some persons worse off without making anyone (non-comparatively) better off. This leveling down would, it seems clear to me, not merely be morally bad, it

¹⁴ The same point applies to benign tumors, of course, though for simplicity we can focus on only one of the two.

would be a change that there is *no* moral reason to pursue. A plausible explanation of what has gone wrong is that PC fetishizes classifications, ignoring the way the value of a classification can vary from person to person and context to context.¹⁵

Could we avoid the problem by further specifying the groups? We might require, for example, that the proportion of *healthy* majority persons (i.e., those with a benign tumor) who receive a benign diagnosis is equal to the proportion of *healthy* minority persons who receive a benign diagnosis. While this would avoid the specific problem in *Medical 2* and focus on the apparently relevant distinction within the groups, all we have done is rediscover one of the specifications of parity of accuracy (which we consider below). Thus, the suggested focus on healthy persons is equivalent to a requirement that ADS have equal true positive rates for the two groups.

What then of parity of accuracy? PA obtains, as we noted above, when the probability of receiving a true classification is (roughly) the same for minority persons and majority persons. Consider:

Medical 3. A hospital employs ADS to determine whether a particular form of cancer is malignant or benign. If untreated, malignant tumors are fatal, but treatment also has very bad side-effects. Accuracy for majority persons is 99%; accuracy for minority persons is 95%.

In *Medical 3*, parity of accuracy does not obtain. Is this morally bad? If it were, then retraining the ADS to achieve or approach PA should in at least one respect be a moral improvement. We can suppose, however, that we cannot improve accuracy for minority persons, so that the only way to approach PA is to reduce accuracy for majority persons. As with PC above, this leveling down would, it seems clear to me, be a wholly awful thing to do since it would result in misdiagnoses of hapless majority persons while making no minority person better off. That cannot be a moral improvement in any respect, and as such, failure to satisfy PA cannot be morally bad.

Consider finally parity of error type. PET obtains, as we noted above, when the probability of a mistaken classification being a false positive (as opposed to a false negative) is (roughly) the same for minority persons and majority persons. PET has seemed appealing to many in the debate on fairness in ADS, specifically in the famous COMPAS controversy, where failure to achieve a form of parity similar to PET was the basis of Pro Publica's charge that the ADS discriminated on the basis of race. However, PET is vulnerable to an objection similar to those raised against PC and PA above. Consider:

Medical 4. A hospital employs ADS to distinguish benign from malignant tumors. If untreated, malignant tumors are fatal for majority persons, but treatment also has very bad side-effects. Untreated malignant tumors are non-fatal for minority persons, although still worse than treatment. The ADS has equal accuracy for the two groups, but makes more false positive errors for majority persons, and more false negative errors for minority persons.

Let us suppose that we could obtain parity of error type by shifting the decision boundary for majority persons only, and let us further suppose (less realistically) that this would not substantially affect accuracy for the group. That is, we retrain the ADS so as to increase the number of false negatives and decrease the number of false positives for majority persons, without changing the overall number of errors. Is there any reason to do so? It strikes me as absurd to suppose that the answer could be "yes", since false negatives are worse than false positives, false negatives are *even worse* for majority persons than false negatives are for minority persons, and no minority person would benefit in any way from the redistribution of errors within the majority group. Once again, if there are situations where we have no reason to promote PET, then failure to satisfy it cannot be what makes indirect algorithmic discrimination morally bad.

It is worth noting that committed egalitarians have defended the view that leveling down can in some cases be better in *one respect*, to wit, in the respect that the outcome is more egalitarian. [84, 98, 107] Does this offer friends of the parity conditions an avenue of response? Not a promising one. In part this is because, although the subject of a complex debate, it is not clear that egalitarian counters are persuasive. [28, 54, 87] More importantly, theoretically developed defenses of egalitarianism do not translate well to the context of parity conditions. Egalitarians are concerned with whole-life distributions of intrinsically valuable goods, such as well-being. One point illustrated by the *Medical* cases above is that classification, accuracy and error type do not correspond to and often do not correlate with such goods. A simple way to illustrate this point is the ease with which we can imagine leveling down scenarios where approaching a parity condition requires increasing whole-life inequality. Egalitarian intuitions will support the parity conditions only when approaching parity promotes equality, e.g., when it makes a disadvantaged minority group better off or a privileged majority group worse off, not when it requires the converse. The parity conditions, however, are symmetrical, that is, they take the moral reasons in favor of the two types of leveling down to be equal. Thus, even

¹⁵ For a related general argument, see [69].

committed egalitarians are likely to find the parity conditions unappealing.¹⁶

I conclude that all three parity conditions are vulnerable to powerful leveling down-type objections. These objections demonstrate that there is no non-contingent moral reason to approach or achieve the parity conditions. This means not just that certain parity conditions may be inappropriate or inapplicable in certain contexts, but that we can and should set parity conditions aside for the purposes of ethical evaluation of an ADS. We will have to look elsewhere for an explanation of what makes indirect algorithmic discrimination morally bad.¹⁷ In the next three sections of the article, I will argue that the harm-based account provides such an explanation.

5 Classification scenarios and harm

Harm-based accounts of the badness of discrimination take point of departure in the claim that causing harm is morally bad [8, 67, 68, 70, 100, 105]. This is perhaps the least controversial claim in moral philosophy—if anything is morally bad, then surely causing harm is—and as such, I will not defend that claim here. If we grant the assumption that causing harm is morally bad, then algorithmic indirect differential effects that cause harm are morally bad, which is to say that it constitutes indirect algorithmic discrimination. Do algorithmic indirect differential effects cause harm? Sometimes, yes. In this and the next two sections, I detail the different ways it can do so, focusing first on differences in the value of classifications across decision scenarios, and next on differences in vulnerability.

Perhaps the most obvious way ADS can cause harm is when the way it classifies a person deprives that person of a good or imposes upon them a bad. This simple observation can explain many of our intuitions about ADS being morally bad. For example, PC may seem appealing if we assume equal base rates in scenarios like *Medical*, because unequal classification then suggests that the ADS misclassifies persons, producing (more) false positives for one group, (more) false negatives for the other, or both. PC loses its attraction in *Medical 2*, where unequal base rates mean that there is no longer any correlation between failure to achieve PC and harmful misclassification. The same point applies to PA and PET in *Medical 3* and *Medical 4* respectively:

while appealing in some scenarios, they lose their appeal when approaching or achieving them no longer correlates with decreasing harm.

There are some important complications to bear in mind, however. The first is that the ways ADS can be bad for a person varies in significant ways across different types of decision scenario. Some decision scenarios distribute goods across true and false classifications, while others distribute across positive and negative classifications. This difference both helps to explain intuitions about some prominent cases and to further illuminate how algorithmic indirect differential effects can cause harm.

In *Medical*, the distribution of goods is across true/false classifications. A false positive is an assessment that mistakenly labels a benign tumor as malignant. A false negative is an assessment that mistakenly labels a malignant tumor as benign. Thus, any *false* classification is bad for the person receiving it. Consider now for comparison:

Criminal. A penal system grants parole to offenders based largely on predicted recidivism, and employs ADS to determine whether offenders are likely to reoffend upon receiving parole. Parole allows offenders to escape the dangerous and oppressive environment of prison to enjoy the benefits and pursue the opportunities of civilian life.

In *Criminal*, a false positive mistakenly labels a low-risk offender as a high-risk recidivist. This is bad for the person because it increases the probability of being denied parole. A false negative mistakenly labels a high-risk offender as a low-risk recidivist. This is good for the person because it increases the probability of being granted parole.¹⁸ Thus, in *Criminal*, the distribution of goods is across positive/negative classifications. Any *positive* classification is bad for the person.

These observations help to explain intuitions in prominent cases. One of the standing concerns in the debate on fairness in ADS is how to prioritize between parity of accuracy and parity of error distribution. In some decision scenarios, the former looks more appealing while in others the latter seems to be more important. We can account for (at least some of) these apparently conflicting intuitions by referencing the difference between *Medical*-type scenarios and *Criminal*-type

¹⁶ A further criticism holds that justice applies at the level of individuals, but parity conditions are concerned with the average treatment of groups. See [18, 69].

¹⁷ The leveling down-type objections apply even if the scenarios are unlikely to occur in practice, but it is worth noting that such scenarios may in fact be common [25, 26].

¹⁸ We set aside here the possibility that it may be all things considered worse for the person likely to reoffend to be granted parole, e.g., because this will allow them to reoffend, and reoffending is bad for the offender. Furthermore, we are still setting aside the issue of when an act, policy or practice might be all-things-considered permissible in spite of the fact that it is bad for some persons, e.g., because denying parole to persons accurately assessed as high-risk recidivists prevents harm to potential victims.

scenarios, specifically the fact that there is a difference with regards to which classifications are bad for persons.

In *Medical*-type scenarios, persons evaluated by the ADS will want accuracy to be as high as possible, and lack of PA means that one group is more likely than the other to be classified in a way that is bad for persons. In *Criminal*-type scenarios, persons evaluated by the ADS will want error distribution to be tilted toward false negatives as much as possible, and lack of PET means that one group is more likely than the other to be mistakenly classified in the way that is bad for the person.

If ADS can classify persons in ways that are bad for them, have we successfully shown that harm can explain algorithmic indirect discrimination? Not quite, for not all algorithmic indirect differential effects that assign a classification that is bad for a person cause harm.

6 Unavoidable bad classifications and harm

What is the relation between classifications that are bad for a person and harm? Consider again *Medical 3*, where the ADS has 99% accuracy for majority persons but only 95% accuracy for minority persons. Because of the way the ADS is trained, it indirectly differentially affects majority and minority persons, in the specific sense that it more often misclassifies minority persons, and misclassifications are bad for persons. This might tempt us to jump immediately to the conclusion that the ADS' indirect differential effects are morally bad because they cause harm. This would be a mistake, however, because there are cases where bad classifications are *unavoidable*, and in such cases the ADS' arguably does not cause harm.

Thus, in *Medical 3*, whether the ADS harms minority persons seems to depend on background conditions that determine *why* parity of accuracy does not obtain. Consider the following three possibilities: the ADS' accuracy for minority persons is (i) pareto-suboptimal, (ii) optimal, or (iii) pareto-optimal.

The ADS' accuracy for minority persons is *pareto-sub-optimal* if the ADS could be retrained with improved accuracy for the minority group and no loss of accuracy for the majority group. In this case, the ADS' indirect differential effects cause harm. There are persons, who will be badly off because they receive the wrong diagnosis, and this is avoidable simply through better training.

However, an algorithm can indirectly differentially affect two groups without the latter condition being met. Lower accuracy for one group may for example be an unavoidable consequence of limitations in data the developers have for training. The accuracy of the algorithm is *optimal* with respect to minority persons if the ADS could not be retrained with improved accuracy for the minority group. In

this case, although the ADS indirectly differentially affects minority persons, in the sense of more frequently misclassifying them, and misclassifications are bad for persons, it is *not* true that minority persons are harmed (in this way) by the indirect differential effects, because there is no way to prevent these misclassifications.¹⁹

The implications of pareto-suboptimality and optimality are simple and relatively uncontroversial. Attention in arguments on fairness and the parity conditions has often been justifiably focused on the third possibility, where the accuracy of the ADS is *pareto-optimal* (but not optimal) with respect to minority persons. That is, the ADS could be retrained with improved accuracy for the minority group, but only at the loss of some accuracy for the majority group. This scenario might occur e.g., if the predictive value of variables differs across groups, and the total accuracy of the ADS is optimized by drawing the decision boundary in a way that more commonly misclassifies minority persons. In some cases, increasing accuracy for one group will lower accuracy for the other group to an even greater extent. In other cases, it may be possible to train the ADS in slightly different ways, that all have the same overall accuracy but redistributes mistakes across groups. Such cases might be rare for certain decision scenarios, but common in others. Consider:

Educational. A university accepts students based on expected academic performance, and employs ADS to rank applicants. There are hundreds of applicants but only the 100 highest ranked applicants are admitted. Education is both a means of valuable self-development and a qualification for many attractive positions on the labor market.

In *Educational*, overestimation of an applicant's academic potential, for example being ranked 71st when one's academic potential really only merits a score of 82nd, is good for the person (in expectation, at least).²⁰ Underestimation

¹⁹ Does it matter what the alternatives to ADS are in the first place, for example how a human doing the same classification task would perform? Yes, clearly. The ADS causes harm if we could do better without it. (cf. [4, 103]. For the purposes of this argument, however, such alternatives ("the human ADS") are no different than the possibility of training a different model. Hence, let us assume that alternatives to ADS are impossible or would be even worse.

²⁰ Overestimation is only *actually* good when it makes a difference to whether the person obtains an education or not. We set aside for simplicity's sake the complex issue of what it means to have academic potential, and whether it can plausibly be ranked. That is, we assume for the purposes of the argument that we can meaningfully speak of a rank that one really merits. Furthermore, as in *Criminal*, we set aside here the possibility that some persons may be *worse* off by being overestimated, e.g., because they are offered and accept a position at an education they are incapable of completing, and the resulting

of an applicant's academic potential is bad for the person for similar reasons. As in *Criminal*, the valence of overestimation (false positives) and underestimation (false negative) differs, since any low rank (negative classification) is bad for the person. However, in *Medical* and *Criminal*, the distribution of goods to any individual is independent of how other persons are classified. In *Educational*, predicted potential is a *positional* good: there are a limited number of absolute goods (education), which are distributed on the basis of the ADS' ranking.

This has two important implications. First, the ADS' treatment of each person directly affects others. Because ranking is a positional good, it can be good for a person that *other* persons receive low rankings (negative classifications), and bad for a person that other persons receive high rankings (positive classifications). Second, the ADS' predictions do not affect the amount of absolute goods distributed: there are exactly 100 positions, and the number of applicants refused is determined entirely by the number of total applicants. As such, training of the ADS cannot affect the *number* of predictions that are bad for a person, only the *identity* of the persons receiving these.

In at least some cases, it may therefore simultaneously be true that some persons are worse off because of the way the ADS classifies them, and false that the ADS or its indirect differential effects is bad because it causes harm. The former because we could retrain the ADS so as to bring it about that some persons received a classification that is good for them; the latter because retraining the ADS would not lead to fewer bad classifications, and so to less harm, but only change the identity of the persons who receive a classification that is bad for them.

This might appear to have counterintuitive implications for the harm-based account of indirect algorithmic discrimination. In such situations, it might seem, we could not condemn indirect differential effects as discriminatory, even if the group of those persons unavoidably harmed is disproportionately composed of minority persons. Let us call this objection:

Musical chairs: In some cases, algorithmic indirect differential effects only change the *identity* of the persons who receive a classification that is bad for a person; they do not change the *number* of persons who receive a classification that is bad for a person. In these cases, the ADS is not bad because it causes harm. But some of these cases are indirect algorithmic discrimination. As such, harm cannot (at least, by

itself) explain what is morally bad about algorithmic indirect discrimination.²¹

The possibility of pareto-optimal indirect differential effects leaves the harm-based account of indirect algorithmic discrimination with an explanatory challenge. In the next and penultimate section, I will argue that we can further elaborate on the harm-based account to explain why it often matters how classifications that are good and bad for a person are distributed across minority and majority persons.

7 Individual and group vulnerabilities

The central argument for the explanatory power of the harm-based account of algorithmic indirect discrimination so far has been to show how the value of classifications can vary across decision scenarios. In addition to this, however, the value of classifications can vary across the persons and groups affected. Some persons and groups are *vulnerable*, and this vulnerability means that they will be more adversely affected by indirect differential effects. I will argue in this section that awareness of such differences in vulnerability has important implications for how we should understand indirect algorithmic discrimination, and that it allows the harm-based account to answer the *Musical chairs* objection, by explaining why indirect differential effects that disadvantage some persons and groups matters more than indirect differential effects that afflict others.

In the above, I have argued that one important reason why indirect algorithmic differential effects can be morally bad is that it may cause harm. However, we have implicitly assumed (for the most part) that the value or disvalue of a classification is the same for all persons.²² This assumption is likely to be false in many cases. Persons will differ in their vulnerability, understood as the degree to which they are harmed by receiving a classification that is bad for a person. Such individual vulnerability will in many cases correlate with membership of minority and majority groups (cf. [4]). We encountered a toy example of this in *Medical 4*, where malignant tumors affected one group more severely than the other, making false negatives worse for majority persons than for minority persons. Minority/majority group correlated differences in vulnerability occur in many real-life

Footnote 20 (continued)

waste of time and experience of failure leave them worse off than they would have been, had they not been offered a position at all.

²¹ Note that the objection does not purport to show that harm explains the badness of *no* cases of indirect algorithmic discrimination. In fact, it is compatible with the objection that harm explains the badness of many cases. The objection is an argument for the more modest claim that harm cannot explain the badness of *all* cases, and that there must therefore be other moral factors at stake.

²² As [81] observe, this dubious assumption is common in both development of ADS and academic discussions of fairness in machine learning.

scenarios e.g., because of the correlation between groups and socio-economic status on the one hand, and socio-economic status and vulnerability on the other.

This simple observation goes a long way toward answering the *Musical chairs* objection. The persons and groups we tend to focus on in the context of indirect discrimination are typically vulnerable, in many cases because of past marginalization and discrimination. This vulnerability means that minority persons tend to suffer greater harm than majority persons from receiving the same (bad) classification. This in turn explains why an ADS may cause harm by distributing classifications that are bad for a person from majority to minority persons, even when there is no change in the number of classifications that are bad for a person.²³ Developers who wish to avoid such harm must take account of individual vulnerability. It seems clear that we ought to do so in *Medical 4*—it is more important to avoid the false negatives that kill patients, than the false negatives that only make patients very ill (bad as that may be)—but exactly the same point applies across all of the contexts where persons differ in vulnerability.

What about group vulnerability? The vulnerabilities I have discussed above remain individual vulnerabilities, even if they correlate with membership of minority groups. Furthermore, there is a sense in which the harms we have been discussing are not really harms that occur *because* of discrimination. It is not *differential* treatment or *differential* effects that causes harm, but the bad classifications and individual vulnerabilities by themselves, as evinced by the fact that in cases like *Medical 3* we could achieve equality by reducing accuracy for majority persons without thereby reducing the harm minority persons suffer.

There are harms, however, that can follow specifically from differential treatment or effects. The most familiar examples of these occur in cases of direct discrimination. Consider:

Medical 5. A hospital performs diagnoses to determine whether a particular form of cancer is malignant or benign. The hospital employs ADS when diagnosing majority persons, and coin-flipping when diagnosing minority persons. If untreated, malignant tumors are fatal, but treatment also has very bad side-effects.

The diagnostic procedure for minority persons—flipping a coin—will of course produce many more errors than the ADS used for majority persons (if not, the hospital urgently

needs to retrain their ADS). This is at least one reason why the differential treatment in *Medical 5* is bad (i.e., discrimination). However, the differential treatment may also cause harm in a different way. If the differential treatment becomes public knowledge, then minority persons will reasonably feel neglected and humiliated. Some persons, majority as well as minority, may also mistakenly take the differential treatment as evidence that the two groups differ in morally relevant respects, or as support for their negative attitudes toward members of the group. Stereotyping, stigmatization, loss of trust and cooperation, loss of self-esteem, resentment, and discrimination in other contexts are realistic consequences of the awareness of the differential treatment (see e.g., [12, 13, 64, 94, 108], cf. also [55]). Plausibly these effects constitute harms, and as such are harms that follow specifically from the differential treatment. Notably, they are harms that occur because of group vulnerabilities, and which apply to members of groups beyond the individual classified. While these harms are more familiar in the context of direct differential treatment, there is no reason to think that they cannot be caused by indirect differential effects.

Group vulnerabilities are the second and final piece of the puzzle about the ways even pareto-optimal indirect differential effects can cause harm. In combination with awareness of the effect of individual vulnerabilities, we can answer the *Musical chairs* objection, and in combination with our understanding of differences in decision scenarios, we can account for the complex ways in which algorithmic indirect discrimination is bad because it causes harm.

8 Concluding remarks

Over the course of this article, I have attempted to first precisely define a particular issue that arises in the use of automated decision systems (ADS), to wit that ADS can indirectly discriminate. I have sketched how the issue emerges, and explored three parity conditions found in the literature on algorithmic fairness. I then used leveling down-type objections to show that we have no moral reason to satisfy any of the parity conditions. Finally, I have argued that that we can explain the moral badness of this form of discrimination by reference to the fact that it causes harm, once we take proper account of the way harm varies across decision scenarios and the persons and groups subject to the ADS. This harm-based account, I have suggested, is at once theoretically plausible and allows us to explain the intuitive differences we respond to in important cases.

Let me make two brief observations on some limitations of the argument. First, while I hope to have shown that the badness of algorithmic indirect discrimination can be explained with reference to the harm that discrimination causes, I have not shown that harm is the only thing that can

²³ It is also possible, as prioritarianism claims, that the moral value of units of well-being vary with the well-being level of the recipient, or that, as telic egalitarianism claims, increasing inequality in the distribution of goods is morally disvaluable. I am not persuaded by either view, but if they were true, harming persons who are in general worse off would be morally even more bad.

make indirect algorithmic differential effects morally bad. The harm-based account is only one general account of what makes discrimination morally bad.²⁴ My current view, which I cannot develop here, is that the most prominent alternatives to the harm-based account of discrimination are vulnerable to powerful general objections, and will in some cases fit poorly with the context of ADS.²⁵ However, those attracted to alternative accounts should be able to accept the present argument while maintaining that algorithmic indirect discrimination is in at least some situations morally bad for other reasons.

Second, what is the feasibility of employing the analysis I have developed to avoid indirect algorithmic discrimination? If the analysis stands, then training an ADS to respect the ethical demands of appropriate concern for the well-being of persons is a more complex task than has been widely recognized. Training must be sensitive to the way the value of classifications can differ across the true/false and positive/negative divisions, and adjust the objective of avoiding the relevant classifications accordingly. Furthermore, developers must consider how the value of errors can vary from one person to another, particularly across the groups conventionally at the heart of our concern for discrimination. When such differences exist, training of the ADS must take into account these differences in order to minimize harm. This will often require “treatment disparity” of some form, i.e., differential treatment [76]. Finally, the ADS must take into account group vulnerability and its effects, such as stigmatic harm. When more equal treatment has the effect of decreasing the harms resulting from group vulnerability, then developers have reason to promote equality to the point where further decreases in these harms are exceeded e.g., by increased harms resulting from loss of accuracy.

Is it realistic that developers can train ADS that accurately take these factors into account to minimize harm? In many situations, probably not. Developers will typically work under constraints of imperfect information and limited resources. As such, developers will often by necessity train

ADSs that cause at least some harm through algorithmic indirect discrimination.²⁶

The inevitability of some harm should not tempt us toward defeatism. If developers employ the understanding developed in the preceding to train ADS so as to minimize the harm done to the best of their ability, then not only will they have done all we can reasonably demand, they will also potentially have protected many vulnerable persons from algorithmic harm. In an imperfect world, doing the least harm one can is often no mean ethical achievement.

Acknowledgements I have presented versions of this paper at the Nordic Network for Political Theory 2019 conference at Oslo University, as well as at seminars at the Food & Health Science department of Copenhagen University, Denmark, the Philosophy department at the Nanyang Technological University, Singapore, the Philosophy & Science Studies department of Roskilde University, Denmark, and the Centre for Experimental-Philosophical Studies of Discrimination (CEPDISC) of Aarhus University, Denmark. I am grateful for helpful questions and comments on these occasions from Andreas Brøgger Albertsen, Kim Angell, Ludvig Beckman, Reuben Binns, Emil J. Busch, Christina Chuang, Jakob Elster, Marion Goodman, Rune Klingenberg Hansen, Frederik Hjorten, Sune Hannibal Holm, Robert Huseby, Ditte Marie Munch Jurisic, Sune Lægaard, Jakob Thrane Mainz, Viki Møller Lynby Pedersen, Jesper Ryberg, Peter Sandøe, Jørn Sønderholm, Kim Mannemar Sønderkov, Jacob Livingston Slosser, Jørn Sønderholm, Olav Benjamin Vassend, and Søren Sofus Wichmann. I owe particular thanks for very thorough written comments to Sebastian Holmen, Nils Holtug, Søren Flinch Midtgaard, and Thomas Søbirk Petersen. Furthermore, I owe an enormous debt of gratitude to my friend, Associate Professor Tommy Sonne Alstrøm, The Technical University of Denmark, who with admirable patience helped me understand the workings of machine learning and automated decision-making. Finally, I am grateful to the Research Department of the Danish Institute for Human Rights, and in particular its former Head of Research, Hans-Otto Sano, as well as CEPDISC, Aarhus University and its leader, Professor Kasper Lippert-Rasmussen. Much of the work on this article was conducted during my tenure as Senior Researcher with the former and visiting Associate Professor with the latter.

Funding Funding was supported by Danmarks Grundforskningsfond, DNR114.

Data availability Not appropriate.

Declarations

Conflict of interest The author attests that he has no conflict of interest.

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²⁴ The most prominent alternative accounts in the literature explain the badness of discrimination with reference to disrespect or inequality. Proponents of respect-based accounts argue that discrimination involves a failure to treat persons in light of reasons grounded in their moral worth [3, 34, 43, 83, 96], or that it involves treating persons in a way that expresses a demeaning underestimation of their worth [50]. Equality-based accounts hold that discrimination involves a decrease in the well-being or life opportunities of persons who are already disadvantaged through no fault of their own. [63, 95].

²⁵ For example, accounts that rely on the discriminator’s mental state are likely to fit poorly with ADS that does not have mental states [103]. For critical discussion of disrespect-based accounts, see [8, 11, 67, 70, 75, 102]. For critical discussion of the expressive disrespect account, see [7, pp. 91–94], [34, pp. 84–90], [70]. For critical discussion of equality-based accounts, see [70].

²⁶ Recent research on how to develop ADS under constraints sensitive to benefits, welfare and harm, includes [4, 26, 25, 48, 97].

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